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=====	25

QUESTION 2

DO NOT REMOVE THE LINE BELOW MAKE SURE 'eel3135_lab07_comment.m' IS IN THE SAME DIRECTORY AS THIS FILE

```
clear; close all; clc;
type('eel3135_lab07_comment.m')
```

```
%% QUESTION #1 COMMENTING
clear
close all
clc
```

```
%% DEFINE FILTER AND INPUT
N = 100;
n = 0:(N-1);
```

```
% FILTER
b = (1/4)*[1 1 1];
```

```

a = [1 -1 0.5 0.25];
% <-- Answer: What is the numerator of this filter's transfer function?
%  $B(z) = (1/4)(1 + z^{-1} + z^{-2})$ 
% <-- Answer: What is the denominator of this filter's transfer function?
%  $A(z) = 1 - z^{-1} + 0.5z^{-2} + 0.25z^{-3}$ 
% Hint: look a later when we implement the filter with the "filter" command
% (use "help filter" to see what it does)
%

% INPUT 1
x1 = zeros(N,1);
x1(1) = 1;
% <-- Answer: (True or False) x1 is an impulse input? If false,
% describe the input.
% True, x1 is an impulse input.

% INPUT 2
x2 = zeros(N,1);
x2(1:12) = cos(3*pi/2* n(1:12));
x2(13:24) = cos(pi/4* n(13:24));
x2(25:36) = cos(2*pi* n(25:36));
x2(37:48) = cos(3*pi/4* n(37:48));
x2(49:60) = cos(pi* n(49:60));
x2(61:72) = cos(pi/8* n(61:72));
x2(73:84) = cos(pi/2* n(73:84));
% <-- Answer: (True or False) x2 is an single frequency input? If
% false, describe the input.
% False: x2 is a piecewise signal made of multiple cosine segments at
% different frequencies, so it is a multi-frequency input.

%% DEFINE AND PLOT OUTPUT

% OUTPUT 1
y1 = filter(b,a,x1);

% OUTPUT 2
y2 = filter(b,a,x2);

% PLOT THE IMPULSE RESPONSE AND DTFT
figure(1)
subplot(311)
stem(n,x1)
xlabel('Time (Samples)')
ylabel('x_1[n]')
subplot(312)
stem(n,y1)
xlabel('Time (Samples)')
ylabel('y_1[n]')
subplot(313)
pzplot(b,a)
axis equal

% PLOT THE IMPULSE RESPONSE AND DTFT

```

```

figure(2)
subplot(311)
stem(n,x2)
xlabel('Time (Samples)')
ylabel('x_2[n]')
subplot(312)
stem(n,y2)
xlabel('Time (Samples)')
ylabel('y_2[n]')
subplot(313)
pzplot(b,a)
axis equal

% =====
% NOTE: YOU DO NOT NEED TO ADD COMMENTS IN THE CODE BELOW. WE JUST
% NEEDED POLE-ZERO PLOTTING CODE AND THUS WROTE IT.
% =====
function pzplot(b,a)
% PZPLOT(B,A) plots the pole-zero plot for the filter described by
% vectors A and B. The filter is a "Direct Form II Transposed"
% implementation of the standard difference equation:
%
%      a(1)*y(n) = b(1)*x(n) + b(2)*x(n-1) + ... + b(nb+1)*x(n-nb)
%                  - a(2)*y(n-1) - ... - a(na+1)*y(n-na)
%
%
% MODIFY THE POLYNOMIALS TO FIND THE ROOTS
b1 = zeros(max(length(a),length(b)),1); % Need to add zeros to get the
right roots
a1 = zeros(max(length(a),length(b)),1); % Need to add zeros to get the
right roots
b1(1:length(b)) = b;      % New a with all values
a1(1:length(a)) = a;      % New a with all values

% FIND THE ROOTS OF EACH POLYNOMIAL AND PLOT THE LOCATIONS OF THE ROOTS
h1 = plot(real(roots(a1)), imag(roots(a1)));
hold on;
h2 = plot(real(roots(b1)), imag(roots(b1)));
hold off;

% DRAW THE UNIT CIRCLE
circle(0,0,1)

% MAKE THE POLES AND ZEROS X's AND O's
set(h1, 'LineStyle', 'none', 'Marker', 'x', 'MarkerFaceColor','none',
'linewidth', 1.5, 'markersize', 8);
set(h2, 'LineStyle', 'none', 'Marker', 'o', 'MarkerFaceColor','none',
'linewidth', 1.5, 'markersize', 8);
axis equal;

% DRAW VERTICAL AND HORIZONTAL LINES
xminmax = xlim();

```

```

    yminmax = ylim();
    line([xminmax(1) xminmax(2)], [0 0], 'linestyle', ':', 'linewidth', 0.5,
'color', [1 1 1]*.1)
    line([0 0], [yminmax(1) yminmax(2)], 'linestyle', ':', 'linewidth', 0.5,
'color', [1 1 1]*.1)

    % ADD LABELS AND TITLE
    xlabel('Real Part')
    ylabel('Imaginary Part')
    title('Pole-Zero Plot')

end

```

```

function circle(x,y,r)
% CIRCLE(X,Y,R) draws a circle with horizontal center X, vertical center
% Y, and radius R.
%
% ANGLES TO DRAW
ang=0:0.01:2*pi;

% DEFINE LOCATIONS OF CIRCLE
xp=r*cos(ang);
yp=r*sin(ang);

% PLOT CIRCLE
hold on;
plot(x+xp,y+yp, ':', 'linewidth', 0.5, 'color', [1 1 1]*.1);
hold off;

end

```

QUESTION 2: Z-TRANSFORM

2 (a) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```

N = 100;
n = 0:N-1;

b = [0 0 0 0 0.75];
a = [1];

x = zeros(N,1);
x(1) = 1;

h = filter(b,a,x);

figure
subplot(211)

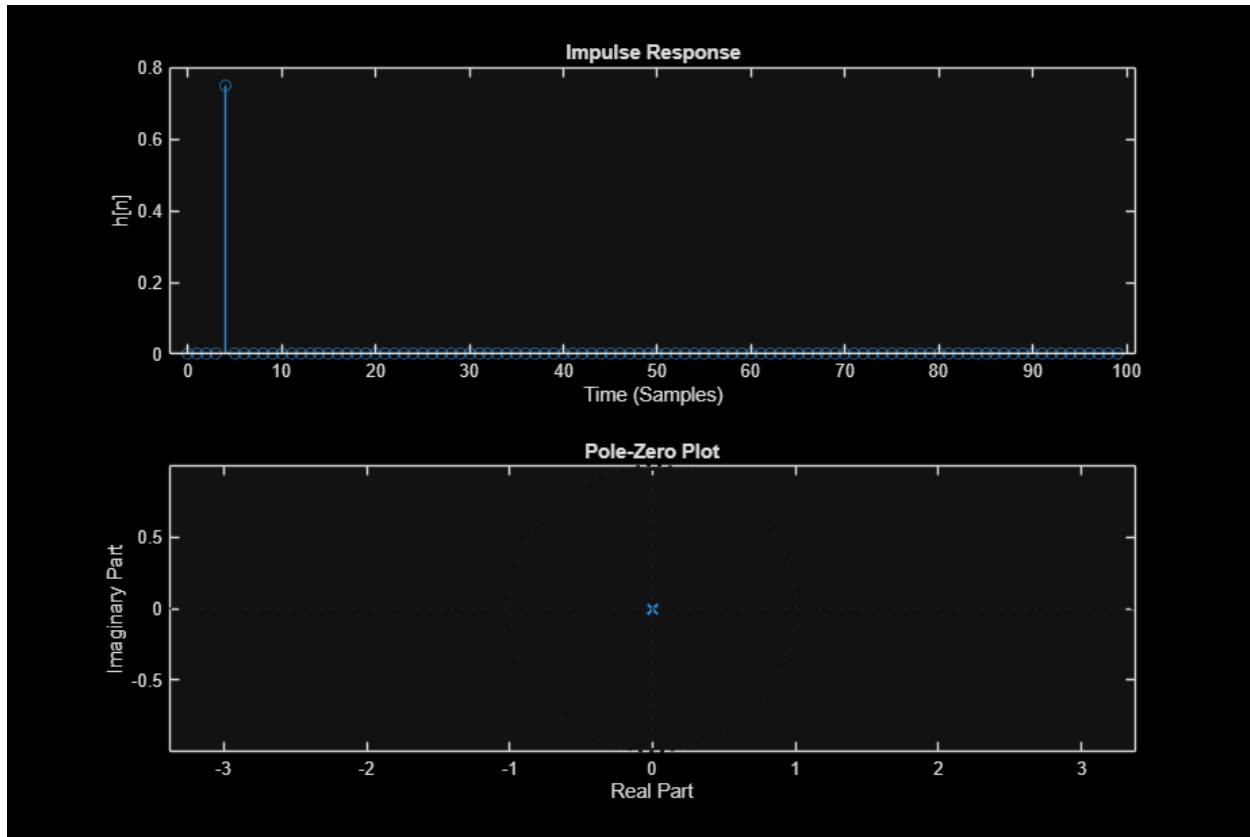
```

```

stem(n,h)
xlabel('Time (Samples)')
ylabel('h[n]')
title('Impulse Response')

subplot(212)
pzplot(b,a)
axis equal

```



2 (b) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```

N = 100;
n = 0:N-1;

b = [1 1];
a = [1];

x = zeros(N,1);
x(1) = 1;

h = filter(b,a,x);

figure
subplot(211)

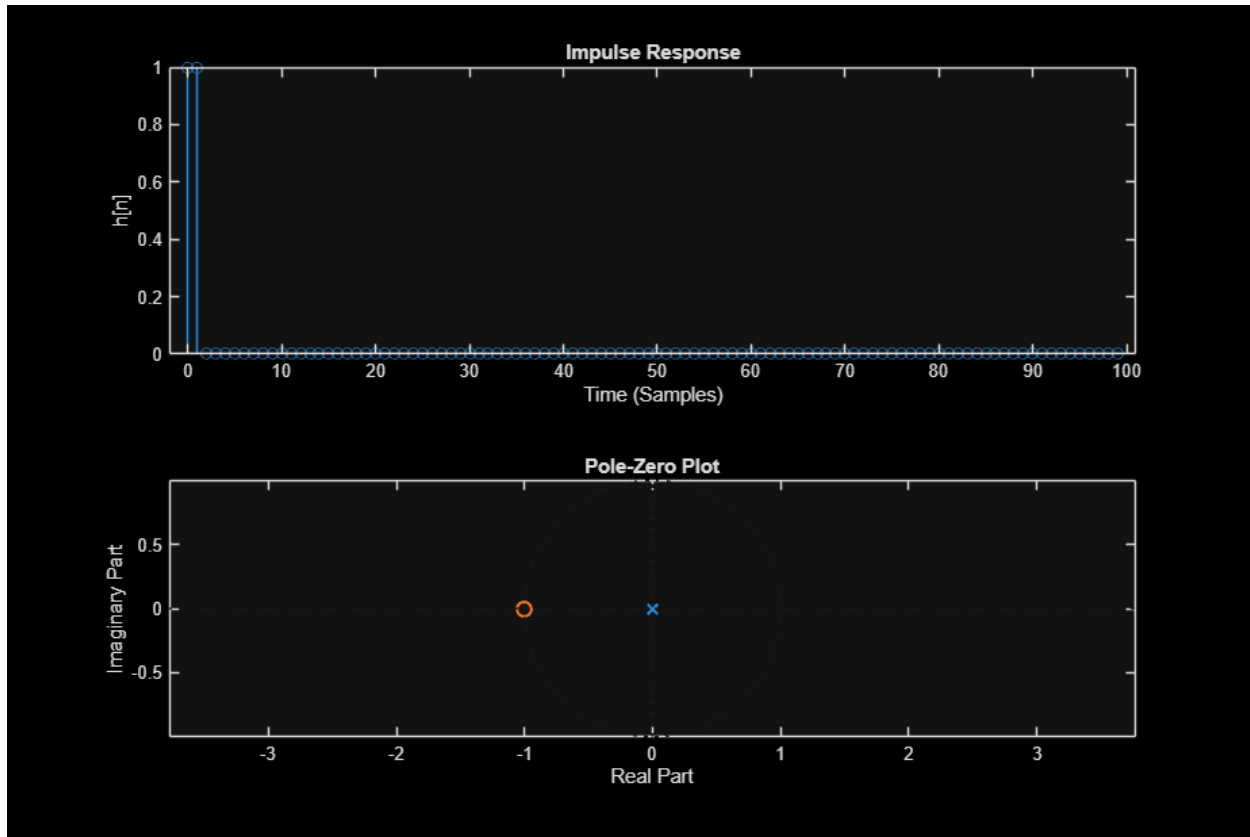
```

```

stem(n,h)
xlabel('Time (Samples)')
ylabel('h[n]')
title('Impulse Response')

subplot(212)
pzplot(b,a)
axis equal

```



2 (c) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```

N = 100;
n = 0:N-1;

b = [1 2 3 4];
a = [1];

x = zeros(N,1);
x(1) = 1;

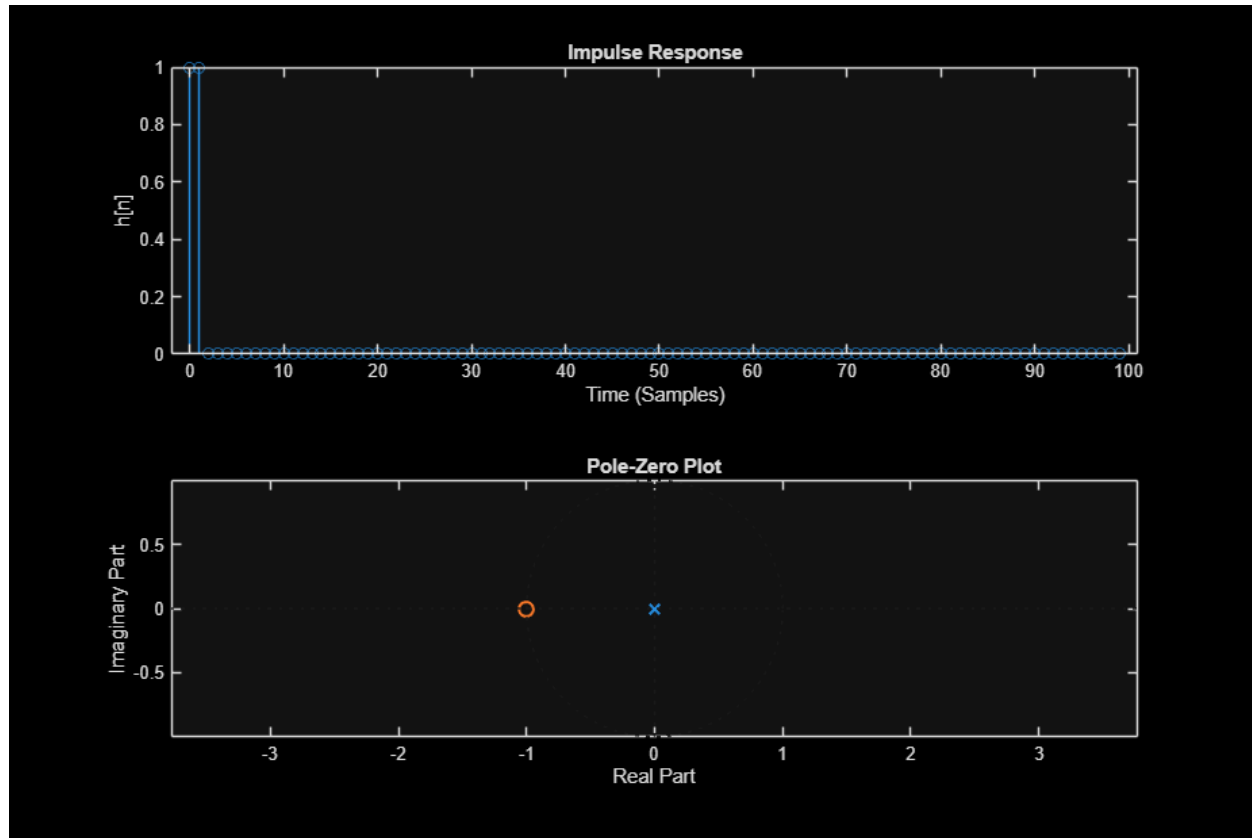
h = filter(b,a,x);

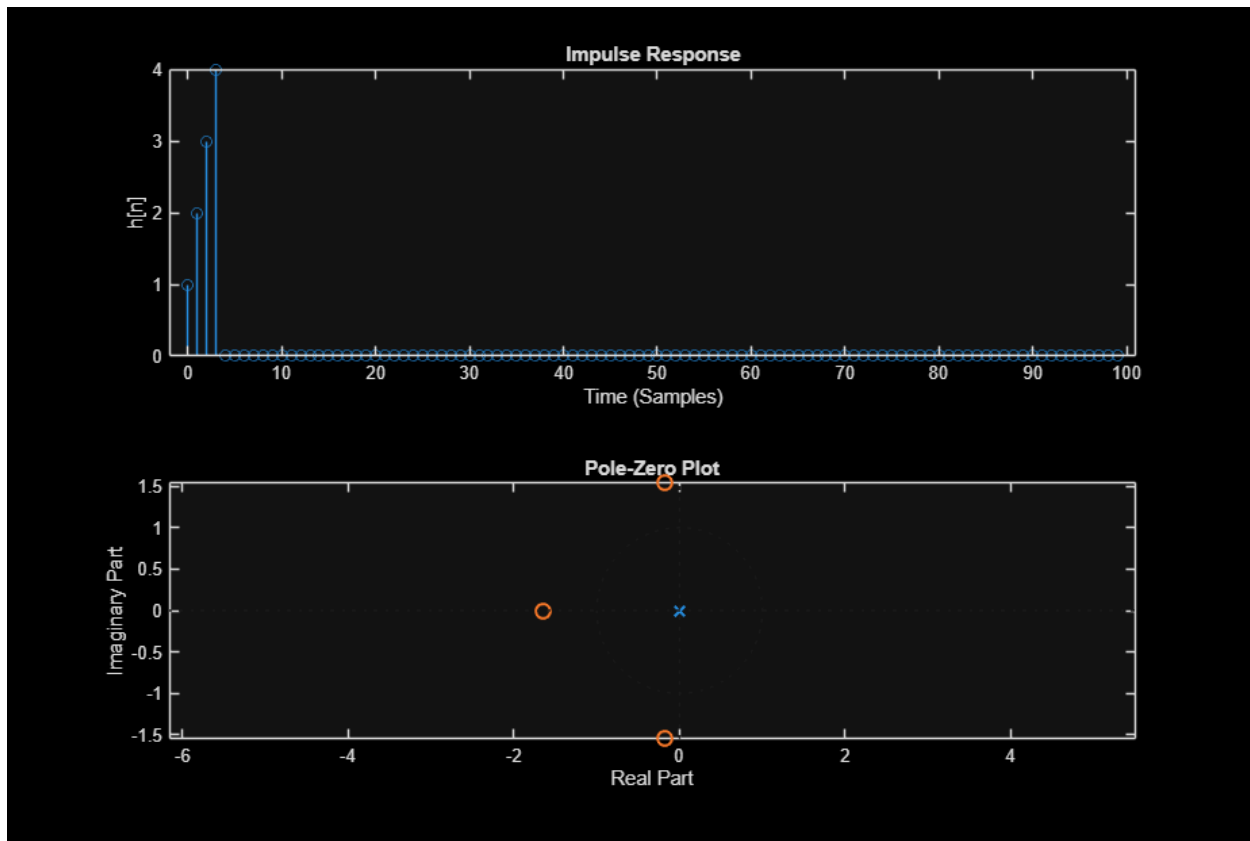
figure
subplot(211)

```

```
stem(n,h)
xlabel('Time (Samples)')
ylabel('h[n]')
title('Impulse Response')

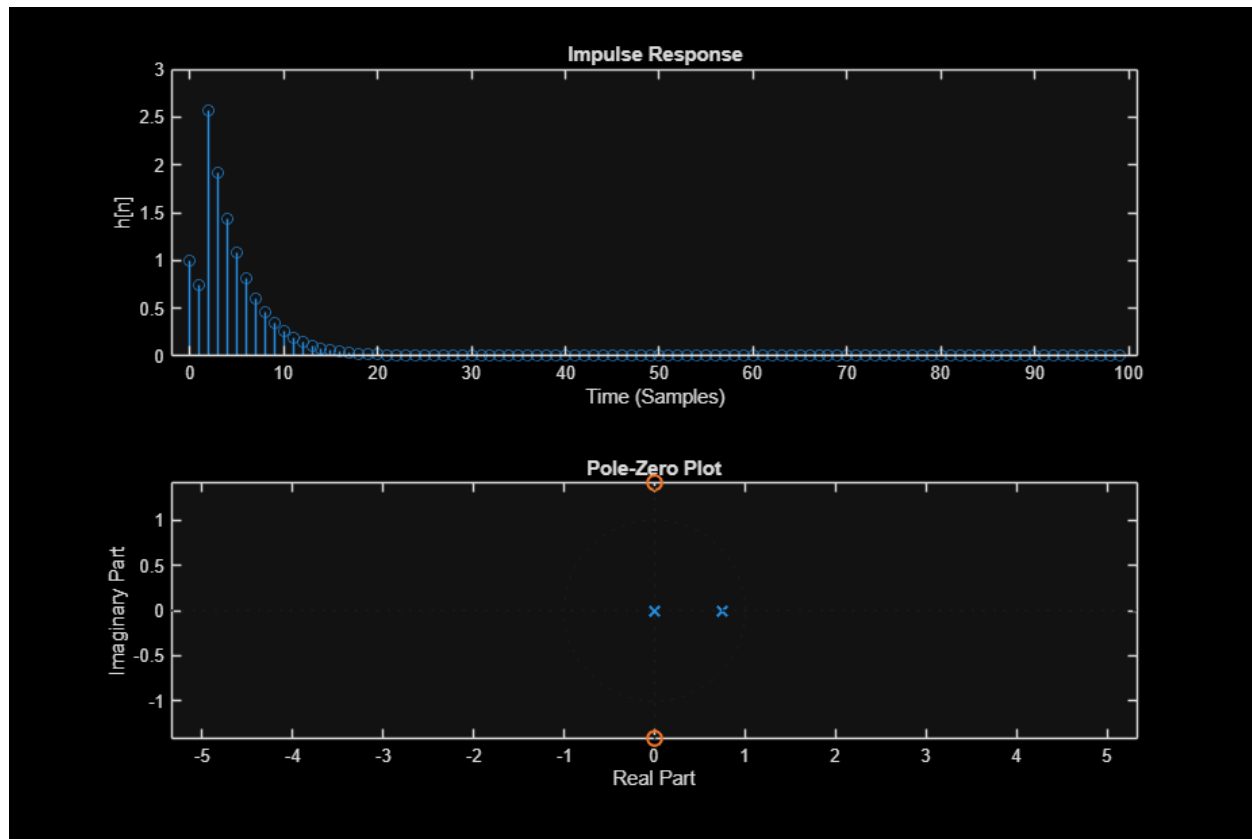
subplot(212)
pzplot(b,a)
axis equal
```





2 (d) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```
N = 100;  
n = 0:N-1;  
  
b = [1 0 2];  
a = [1 -0.75];  
  
x = zeros(N,1);  
x(1) = 1;  
  
h = filter(b,a,x);  
  
figure  
subplot(211)  
stem(n,h)  
xlabel('Time (Samples)')  
ylabel('h[n]')  
title('Impulse Response')  
  
subplot(212)  
pzplot(b,a)  
axis equal
```

2 (e) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```

N = 100;
n = 0:N-1;

b = [1 0 2];
a = [1 1.75];

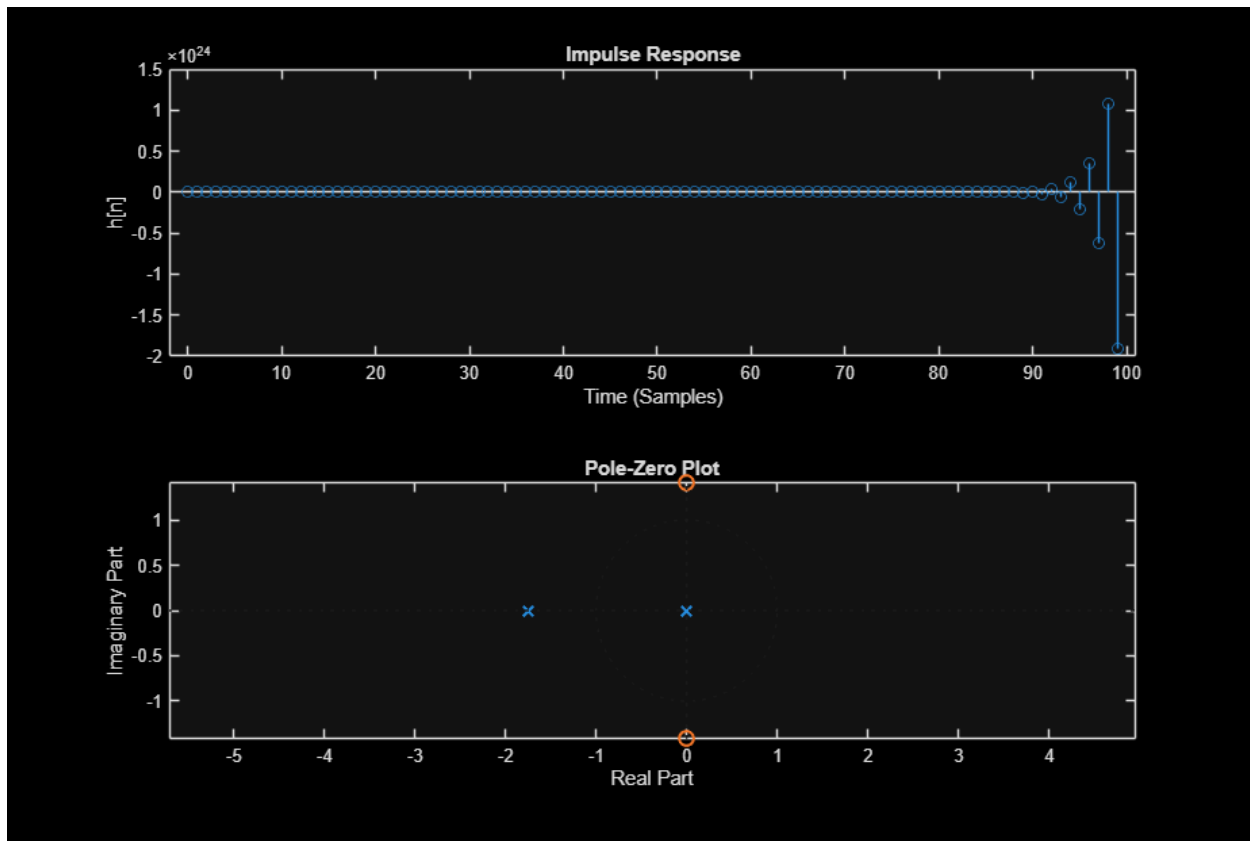
x = zeros(N,1);
x(1) = 1;

h = filter(b,a,x);

figure
subplot(211)
stem(n,h)
xlabel('Time (Samples)')
ylabel('h[n]')
title('Impulse Response')

subplot(212)
pzplot(b,a)
axis equal

```



2 (f) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```

N = 100;
n = 0:N-1;

b = [1];
a = [1 0 -0.85 0 0.7225];

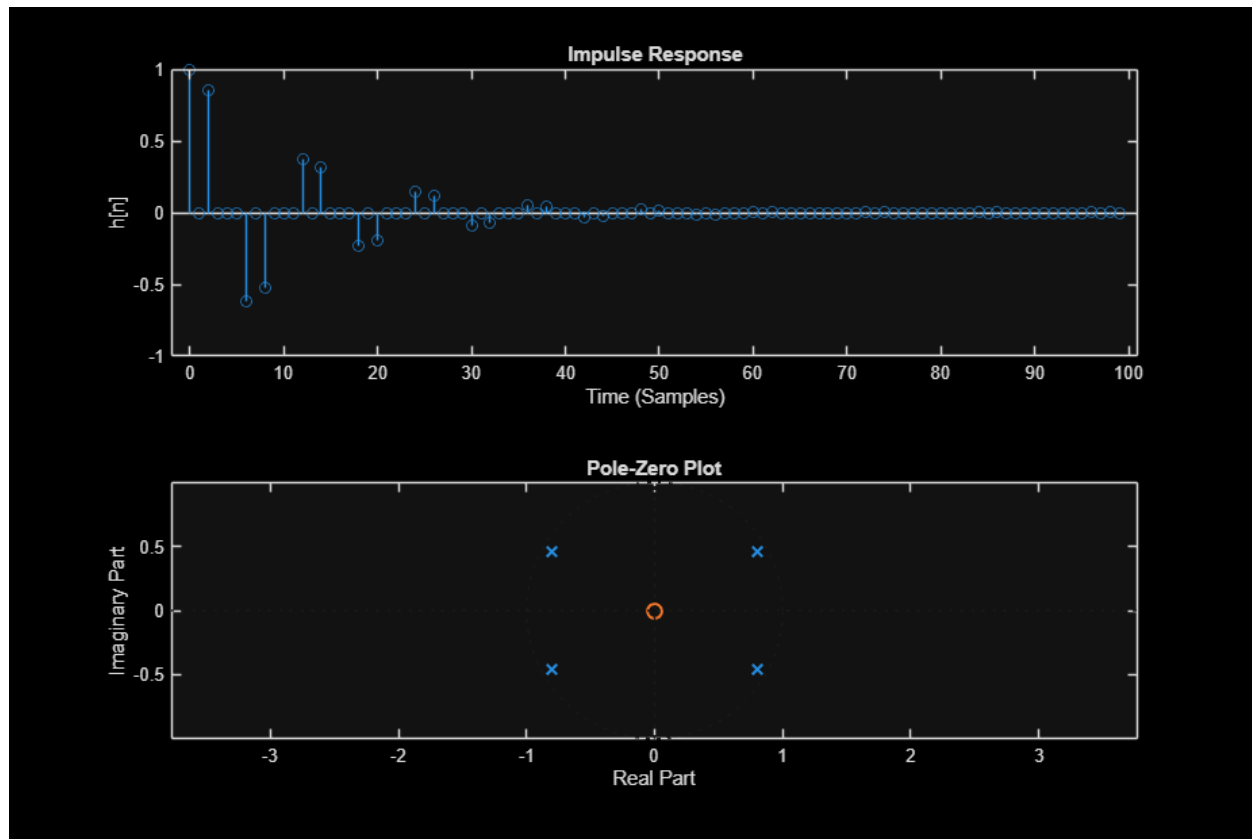
x = zeros(N,1);
x(1) = 1;

h = filter(b,a,x);

figure
subplot(211)
stem(n,h)
xlabel('Time (Samples)')
ylabel('h[n]')
title('Impulse Response')

subplot(212)
pzplot(b,a)
axis equal

```



2 (g) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```

N = 100;
n = 0:N-1;

b = [1 0 0 -1];
a = [1];

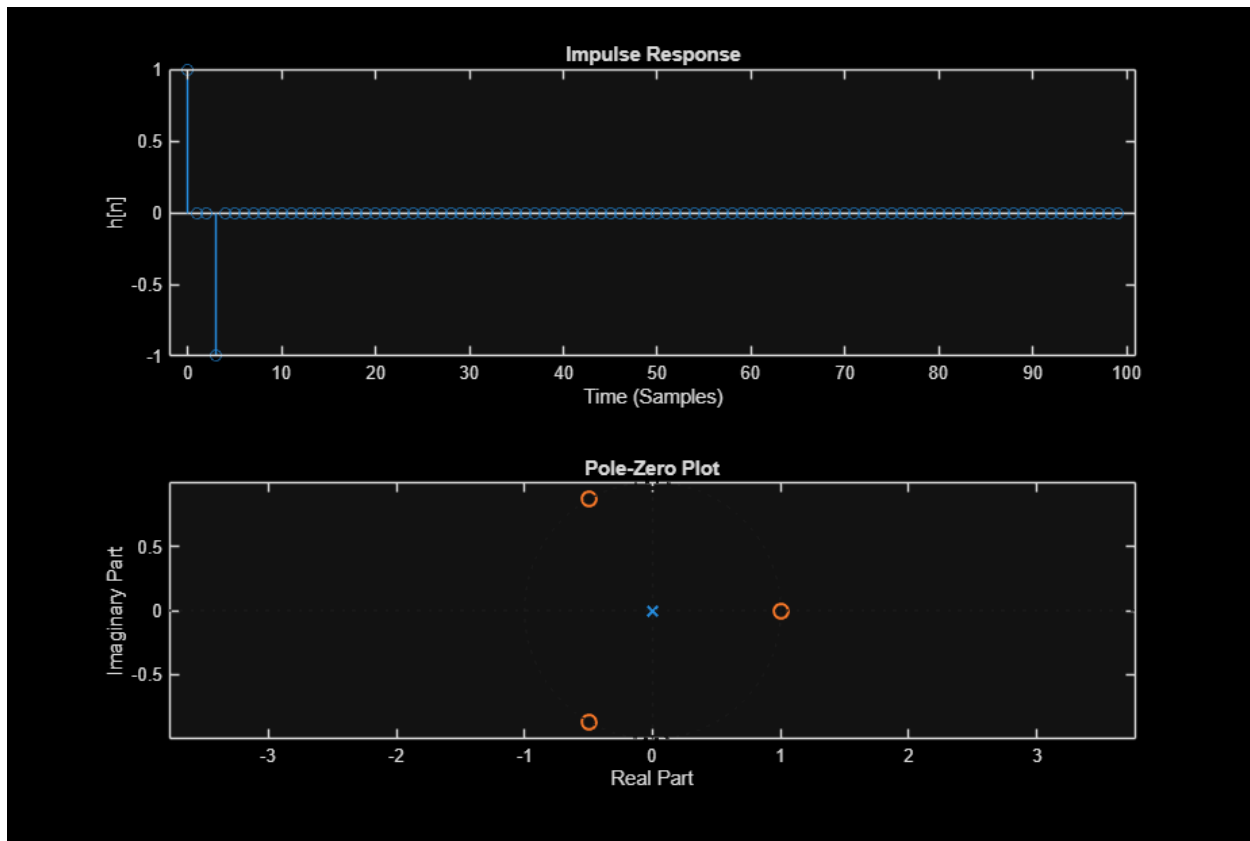
x = zeros(N,1);
x(1) = 1;

h = filter(b,a,x);

figure
subplot(211)
stem(n,h)
xlabel('Time (Samples)')
ylabel('h[n]')
title('Impulse Response')

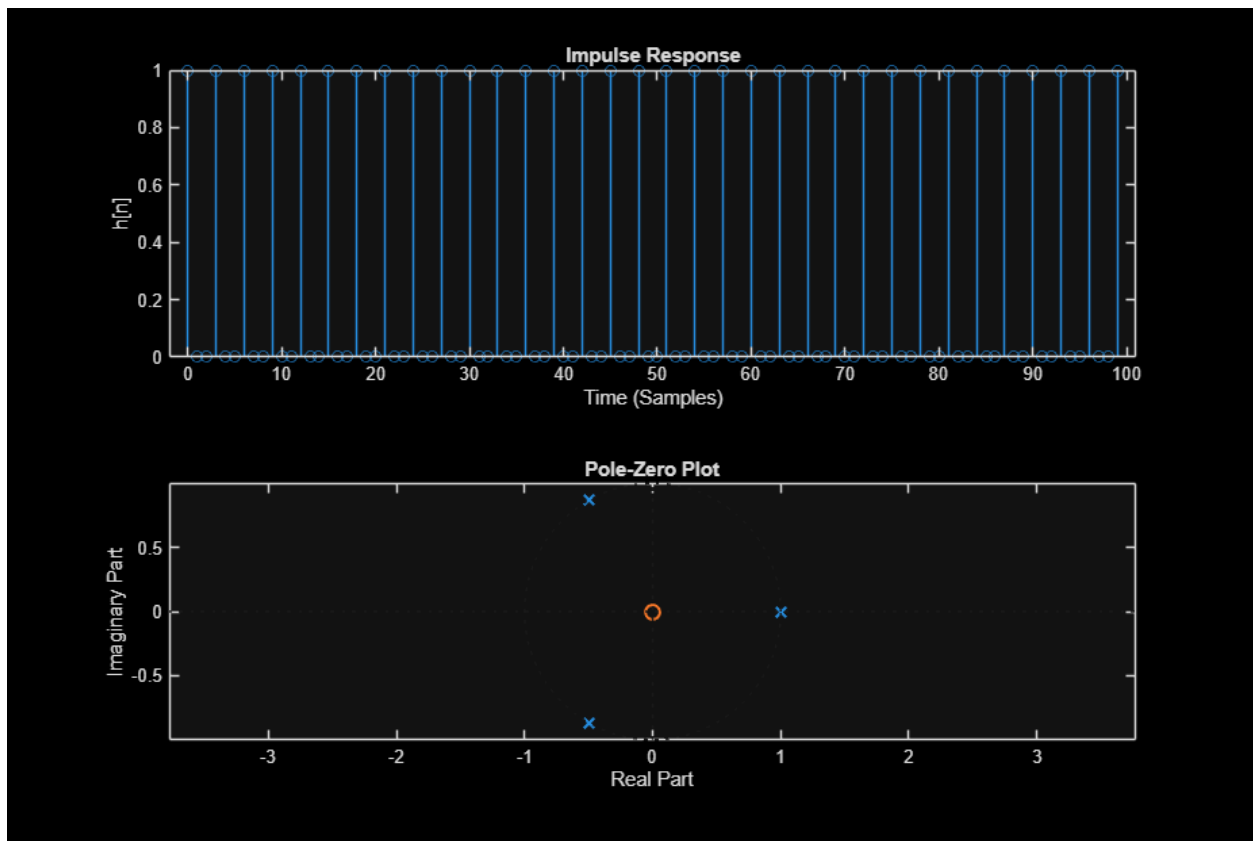
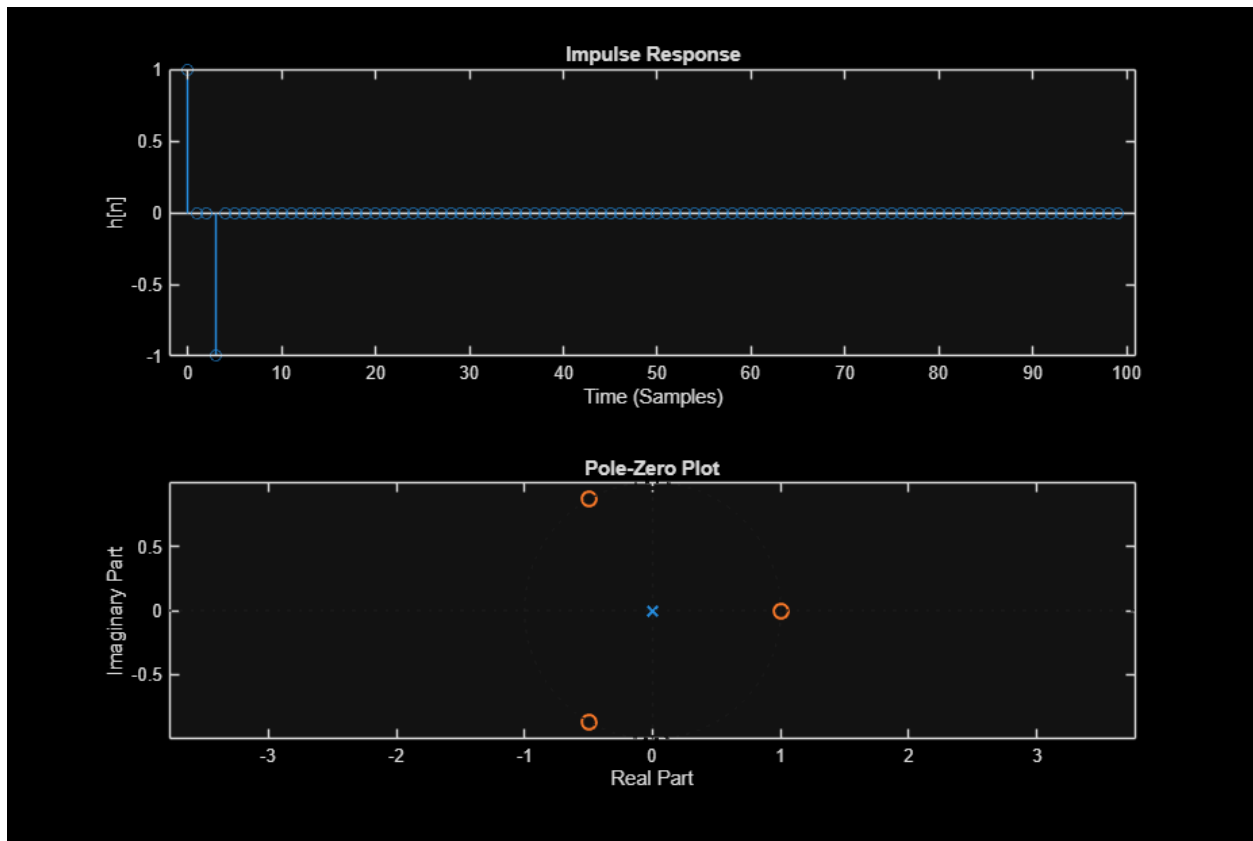
subplot(212)
pzplot(b,a)
axis equal

```



2 (h) PLOT IMPULSE RESPONSE AND POLE-ZERO PLOT

```
N = 100;  
n = 0:N-1;  
  
b = [1];  
a = [1 0 0 -1];  
  
x = zeros(N,1);  
x(1) = 1;  
  
h = filter(b,a,x);  
  
figure  
subplot(211)  
stem(n,h)  
xlabel('Time (Samples)')  
ylabel('h[n]')  
title('Impulse Response')  
  
subplot(212)  
pzplot(b,a)  
axis equal
```



QUESTION 3: MORE Z-TRANSFORM

3 (a) ANSWER QUESTION

```
N = 51;
n = 0:N-1;
alpha = 0.06;

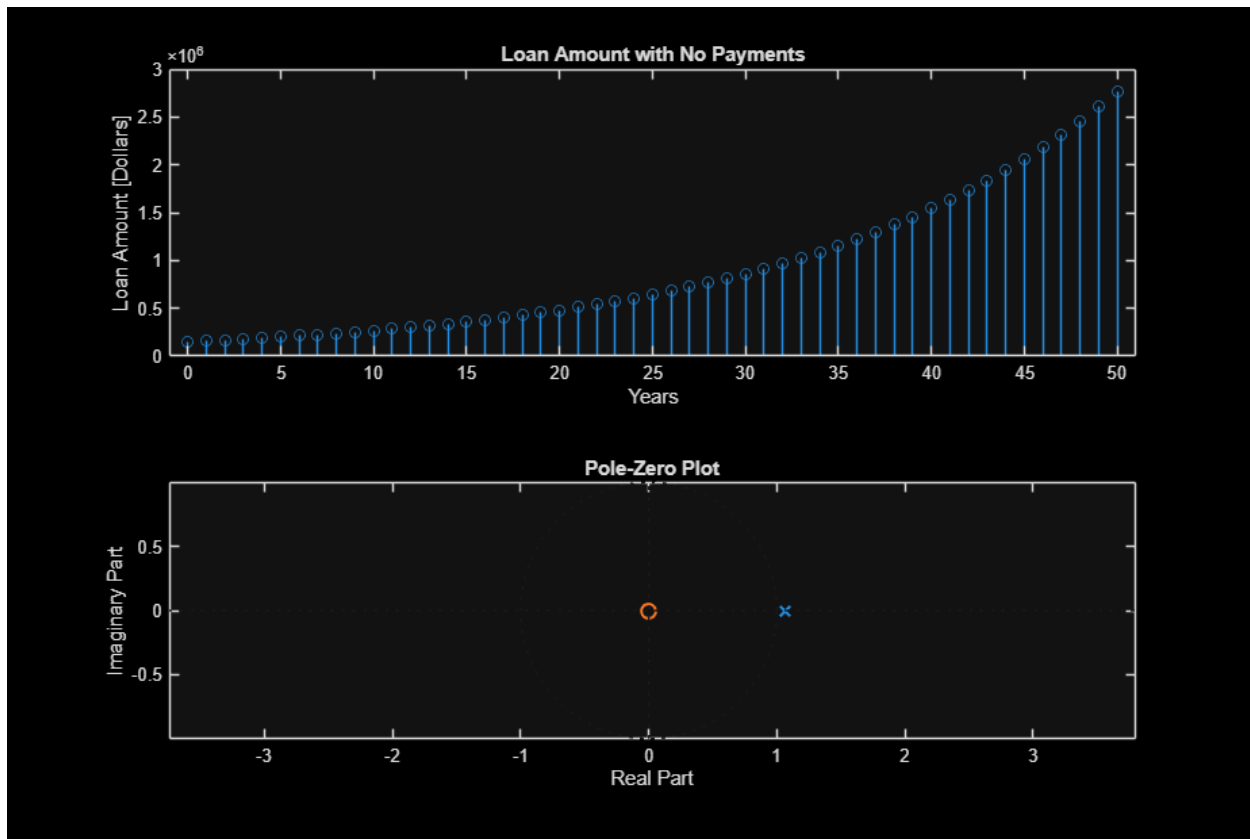
b = [1];
a = [1 -(1+alpha)];

x = zeros(N,1);
x(1) = 150000;

y = filter(b,a,x);

figure
subplot(211)
stem(n,y)
xlabel('Years')
ylabel('Loan Amount [Dollars]')
title('Loan Amount with No Payments')

subplot(212)
pzplot(b,a)
axis equal
```



3 (b) ANSWER QUESTION

For the preceding parameters, it will take about 32.56 years before the loan reaches 1 million dollars, so the loan first exceeds 1 million dollars in year $n = 33$.

3 (c) ANSWER QUESTION

```
N = 51;
n = 0:N-1;
alpha = 0.06;

b = [1];
a = [1 -(1+alpha)];

x = zeros(N,1);

% loan taken at n=0
x(1) = 150000;

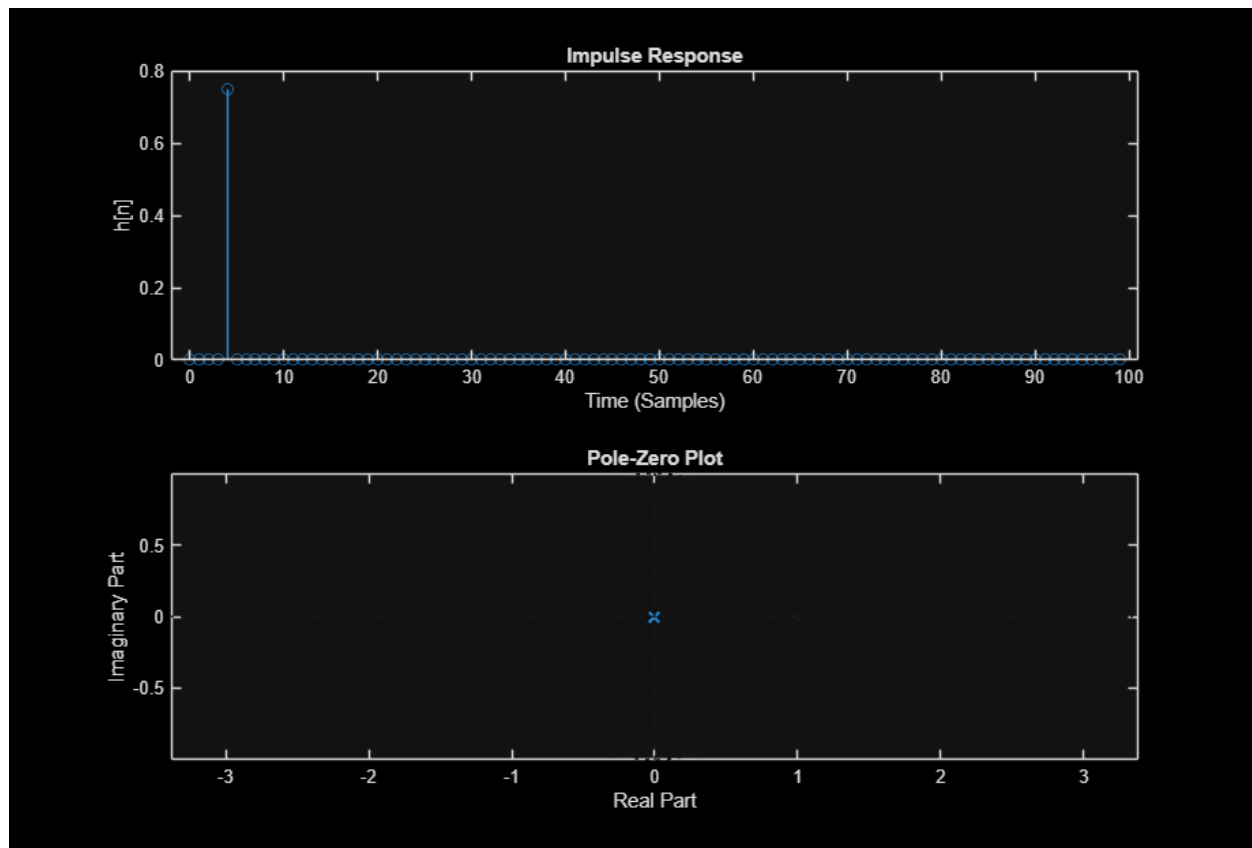
% payments starting year 8
for k = 9:N
    x(k) = -15000;
end

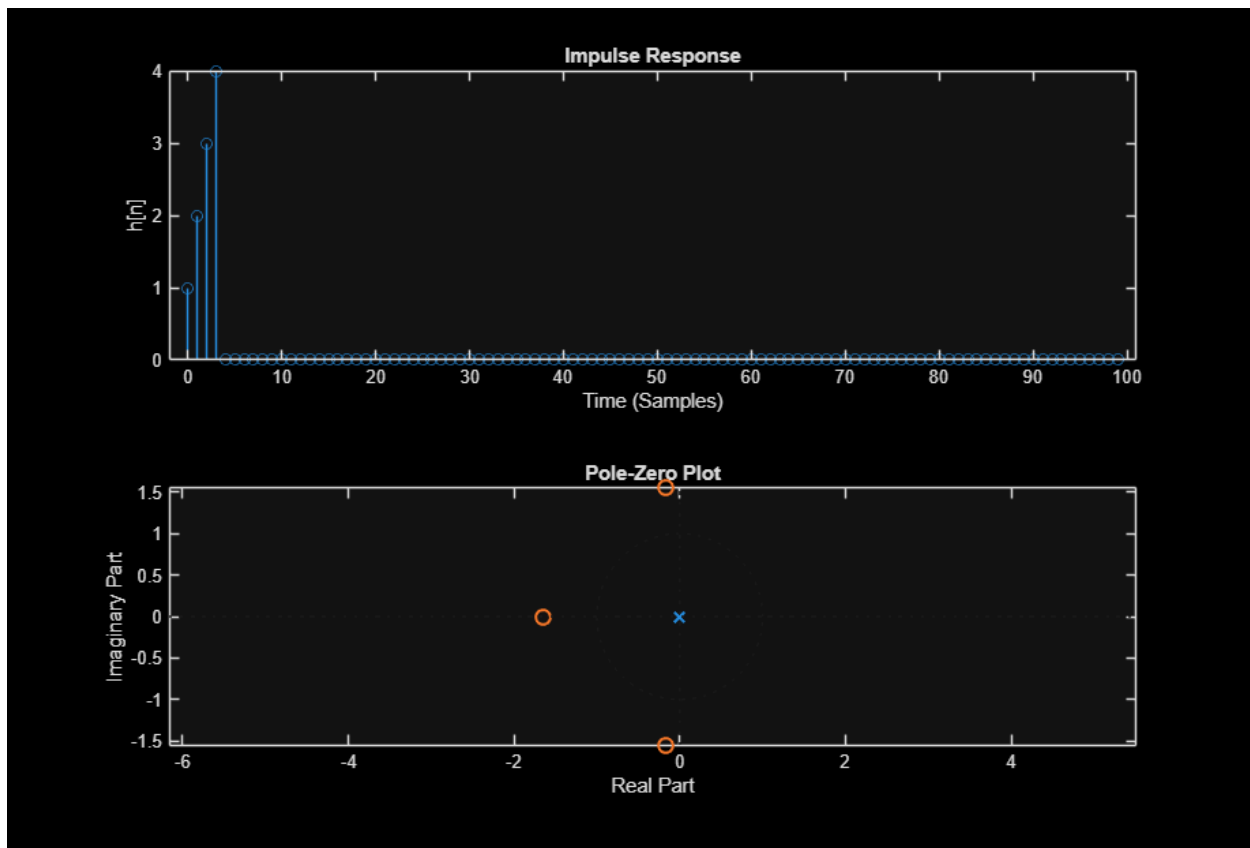
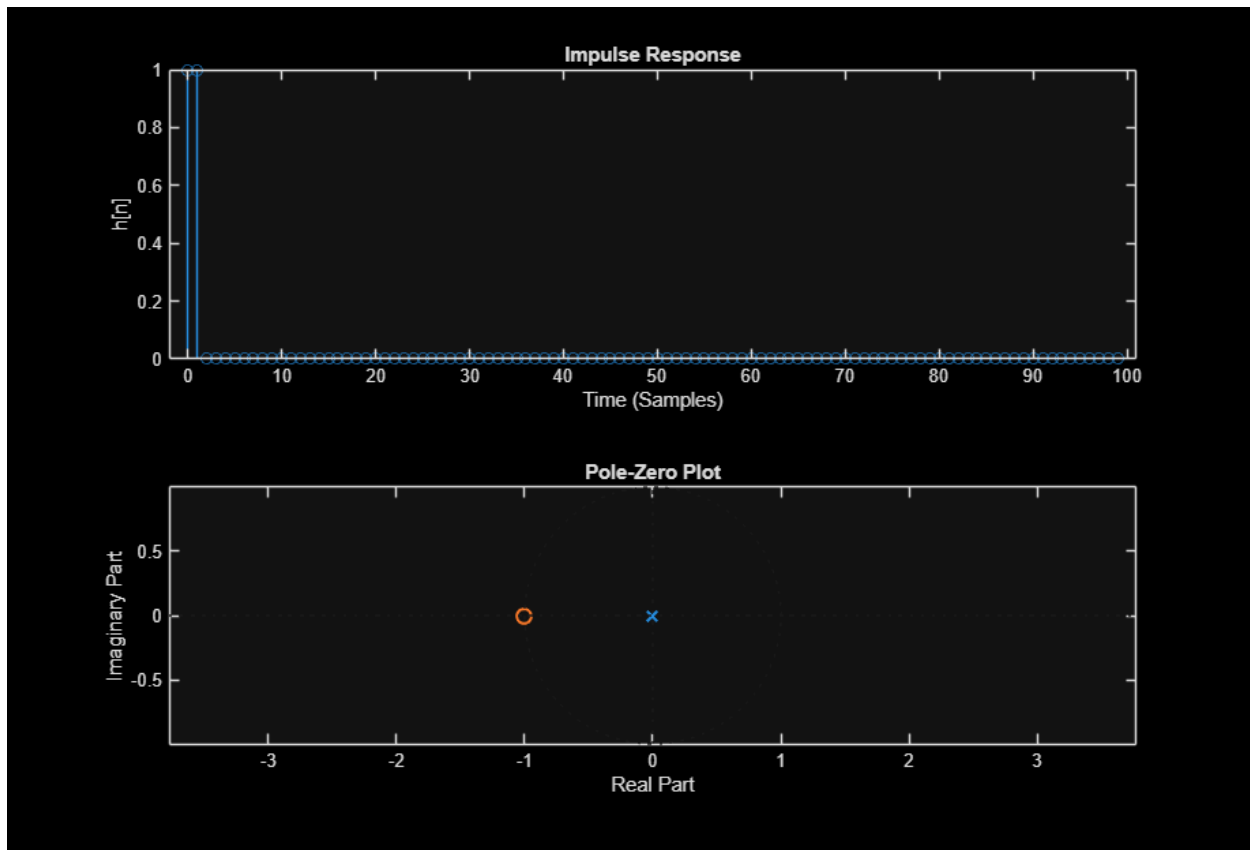
y = filter(b,a,x);
```

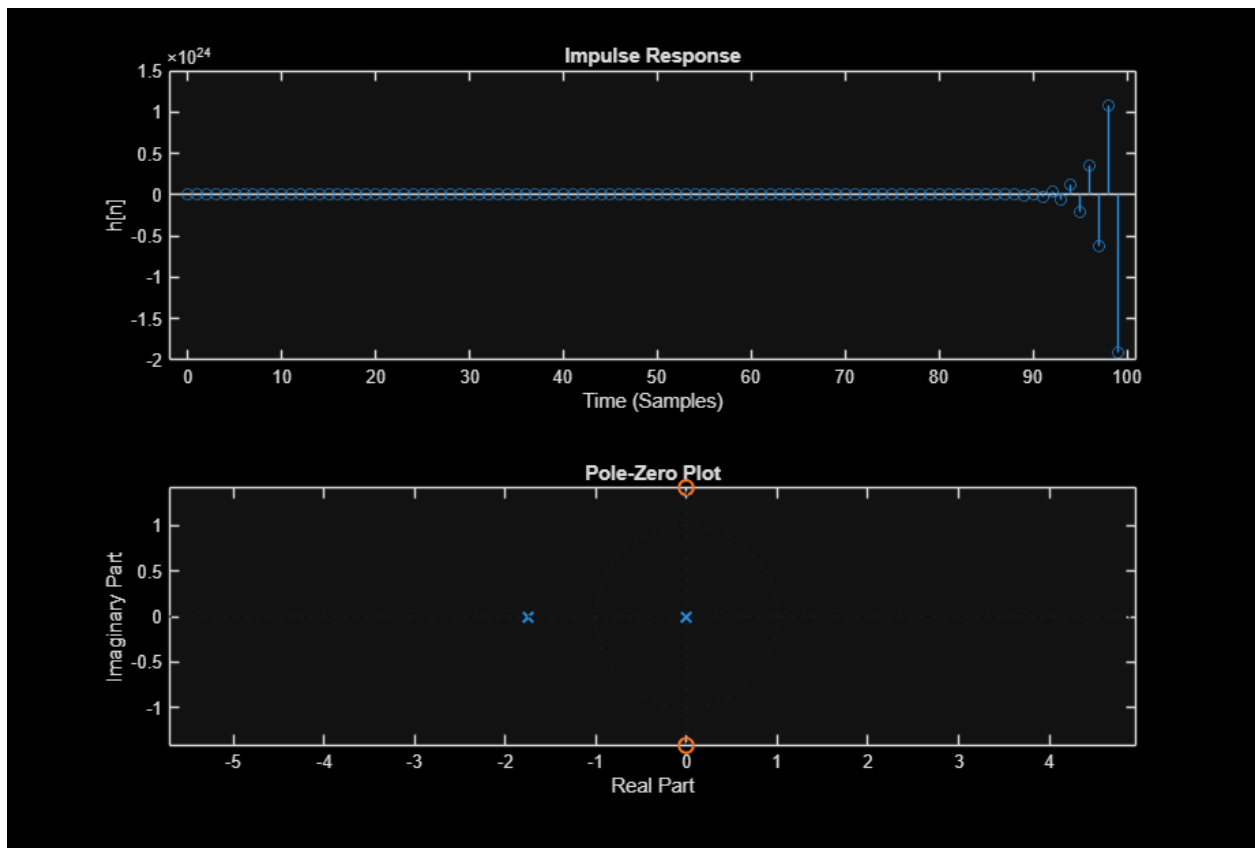
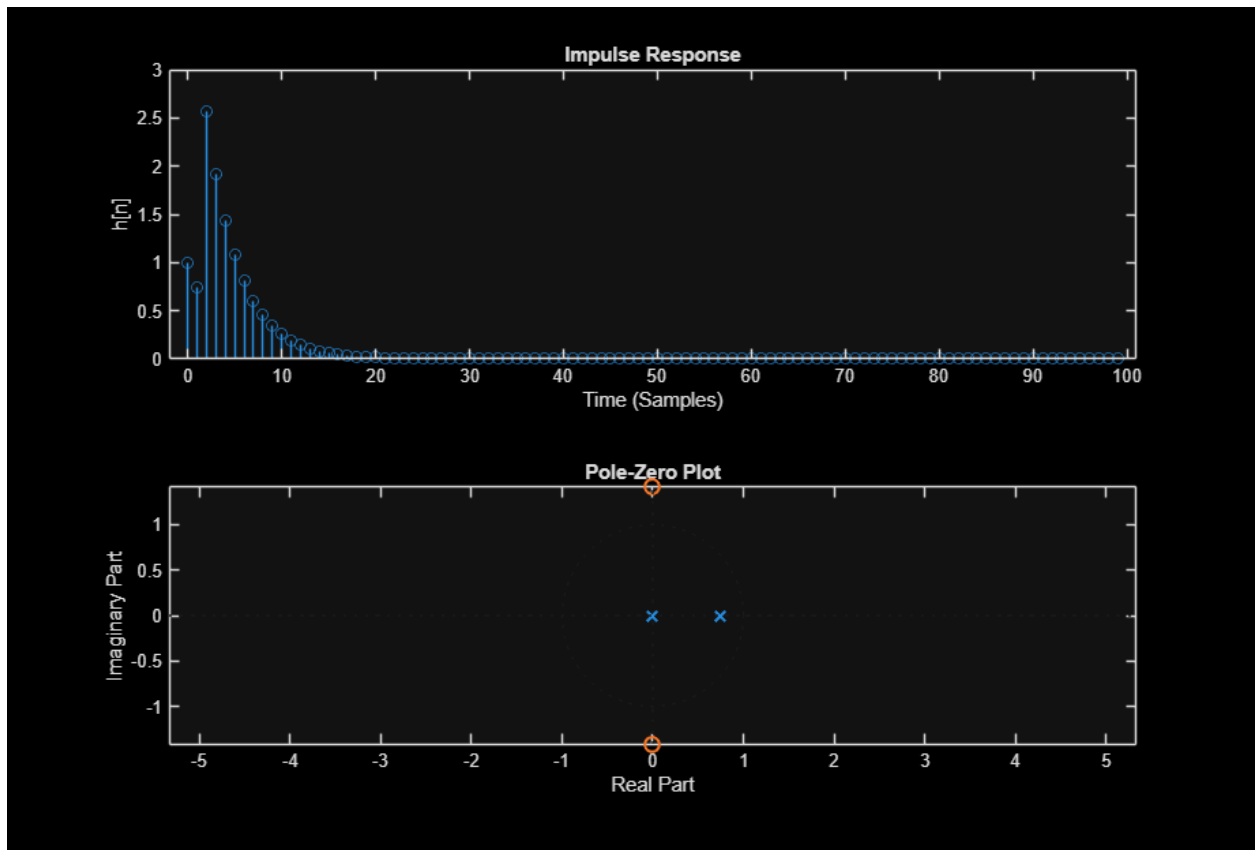
```
figure
```

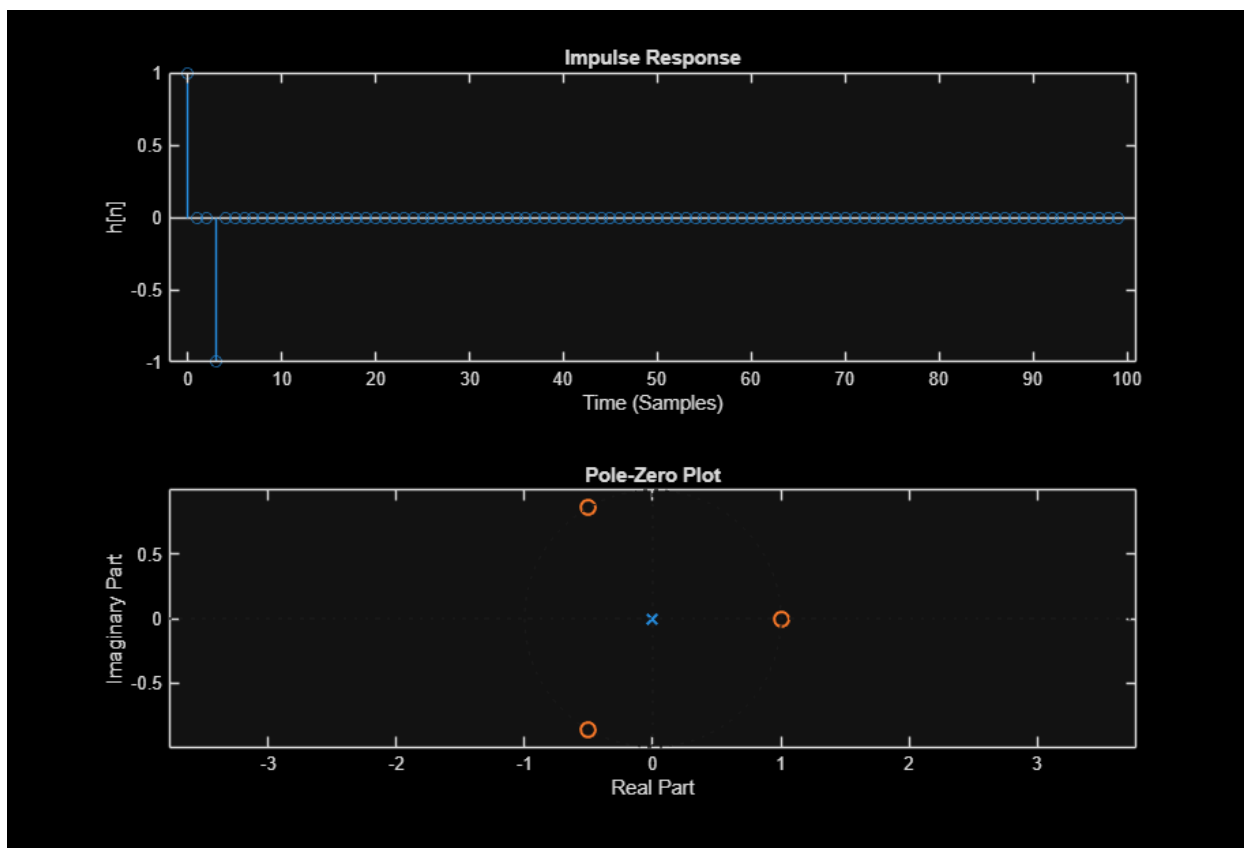
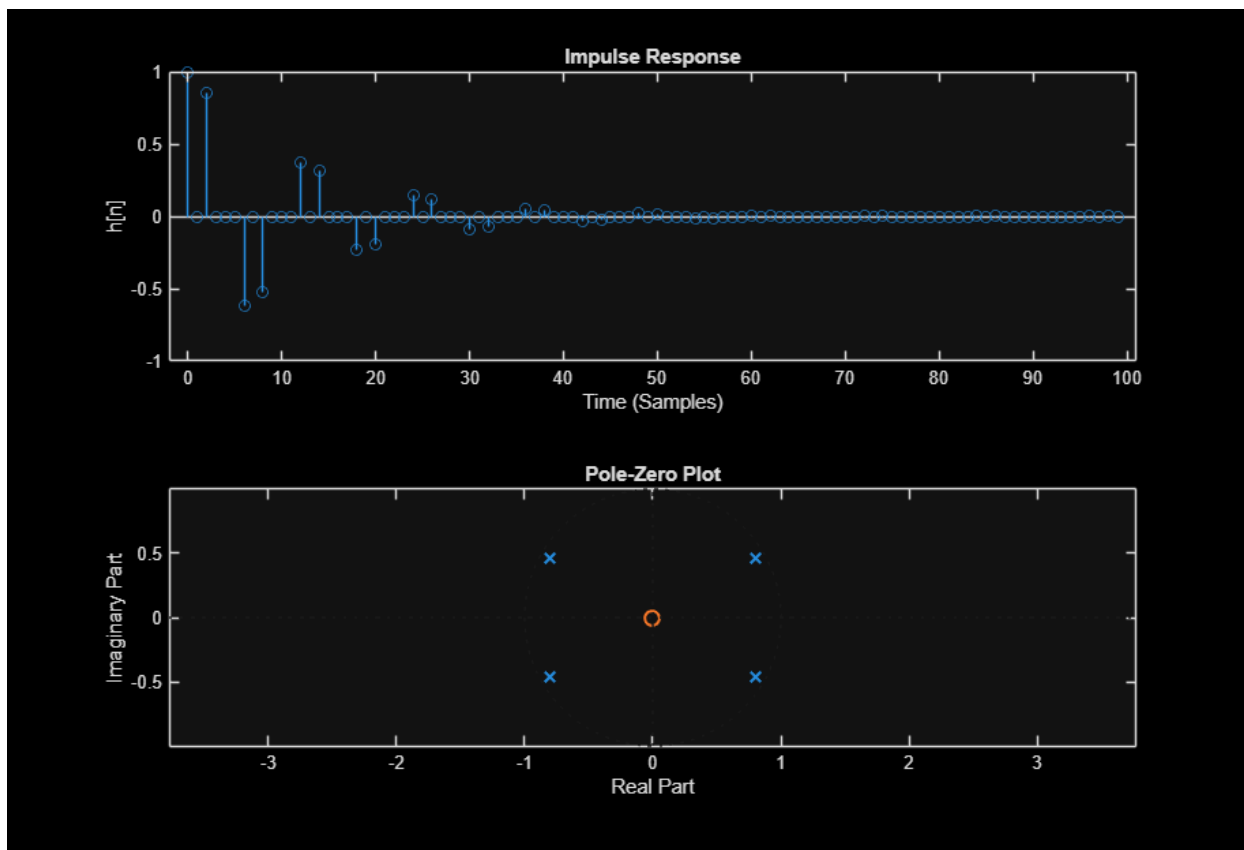
```
subplot(211)
stem(n, max(y,0))
xlabel('Years')
ylabel('Loan Amount [Dollars]')
title('Loan Balance With Payments')
```

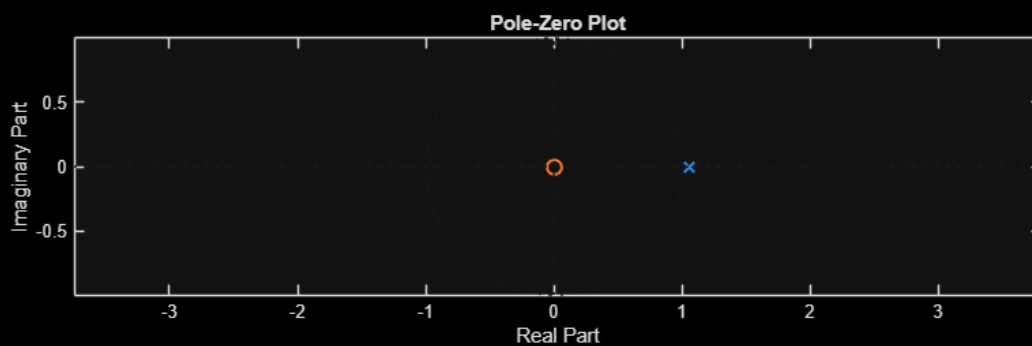
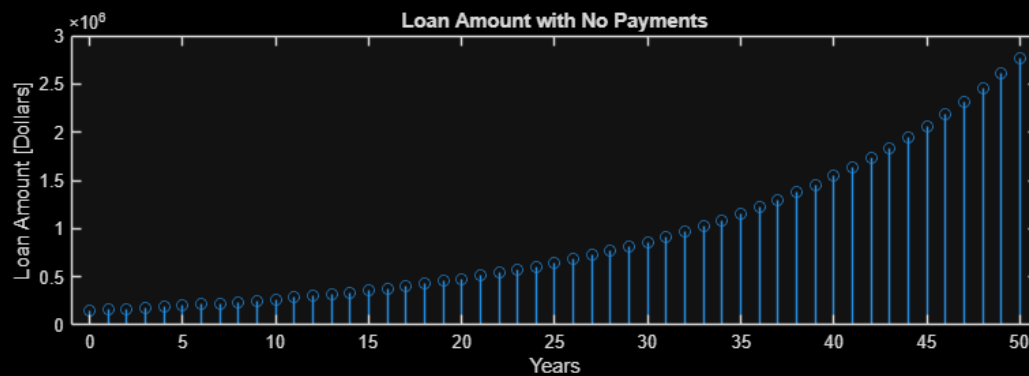
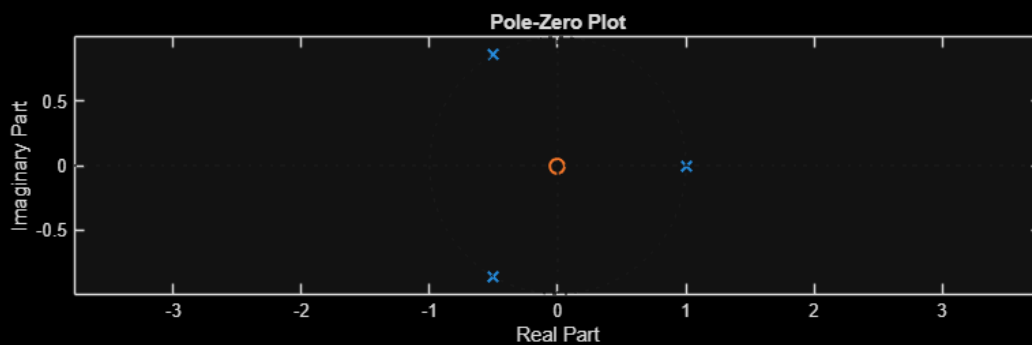
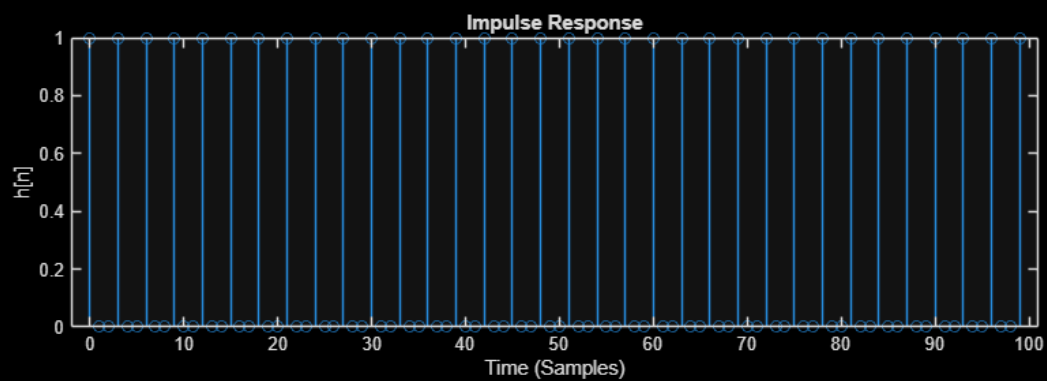
```
subplot(212)
pzplot(b,a)
axis equal
```

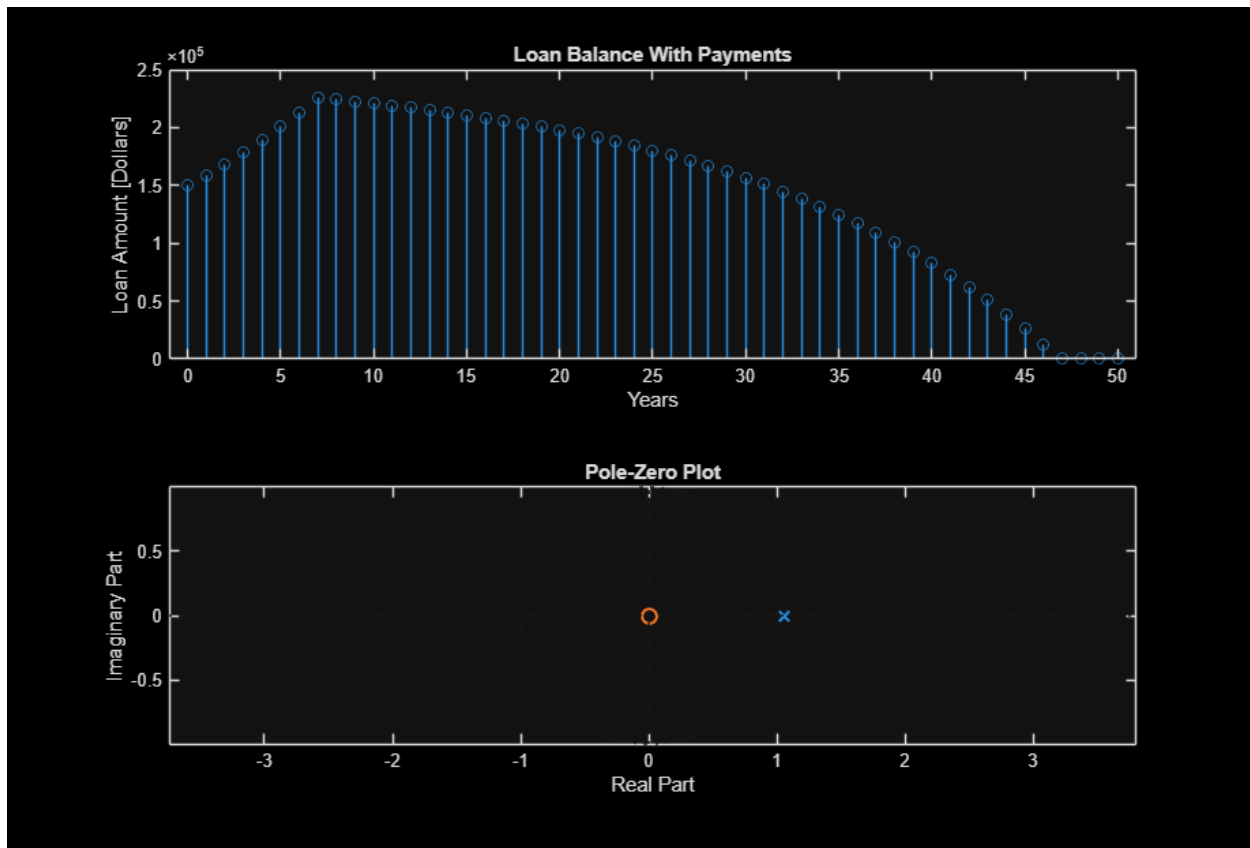












3 (d) ANSWER QUESTION

For the preceding parameters, the loan amount goes to zero. The loan is first less than 10% of the original loan (less than \$15000) at $n = 46$ years.

3 (e) ANSWER QUESTION

In year $n = 0$ you need to start paying to pay the entire loan on or before year $n = 15$. Technically the graph I created below still shows a positive value of 10344.2 at $x = 15$, but this is lower than the payment amount you will make each year. So technically, once you make the payment for that year, year 15 will be the last year you need to make a payment if you start paying off the loan at $n = 0$. It is impossible to start sooner than this to make it payed off before year 15 with the current parameters.

```
%just to verify:
% alpha = 0.06;
% N = 51;
% n = 0:N-1;
%
% b = [1];
% a = [1 -(1+alpha)];
%
% start = 0; %to test different start years
```

```
%
% x = zeros(N,1);
%
% % initial loan at n = 0
% x(1) = 150000;
%
% % payments begin after chosen start year
% for k = start+2:N
%     x(k) = -15000;
% end
%
% y = filter(b,a,x);
%
% figure
% stem(n,max(y,0))
% xlabel('Years')
% ylabel('Loan Amount [Dollars]')
% title(['Loan Balance with Payments starting at year ', num2str(start)])
```

3 (f) ANSWER QUESTION

```
N = 51;
n = 0:N-1;

alpha = 0.06;
gamma = 0.1;
Q = 8;

a = [1 -(1 + alpha - gamma)];

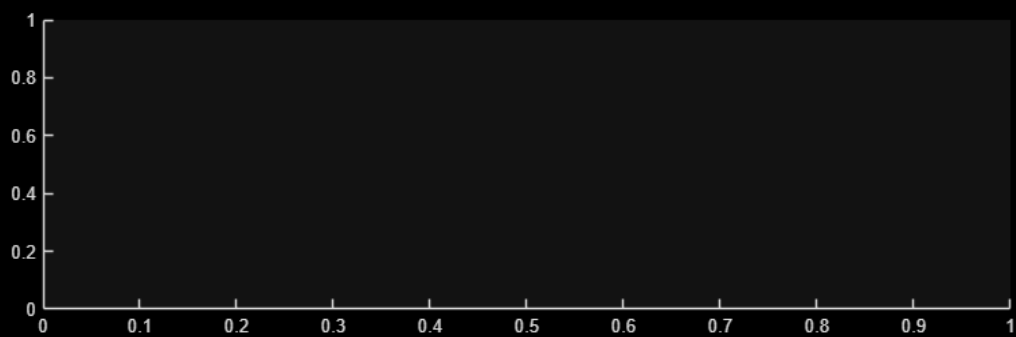
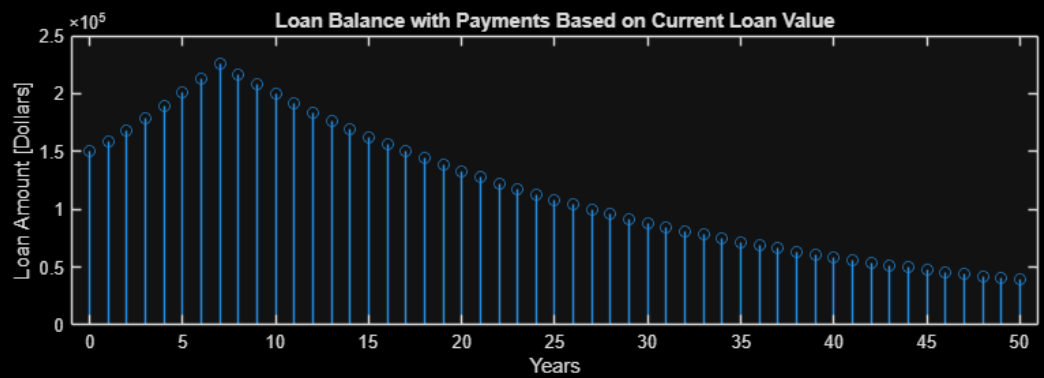
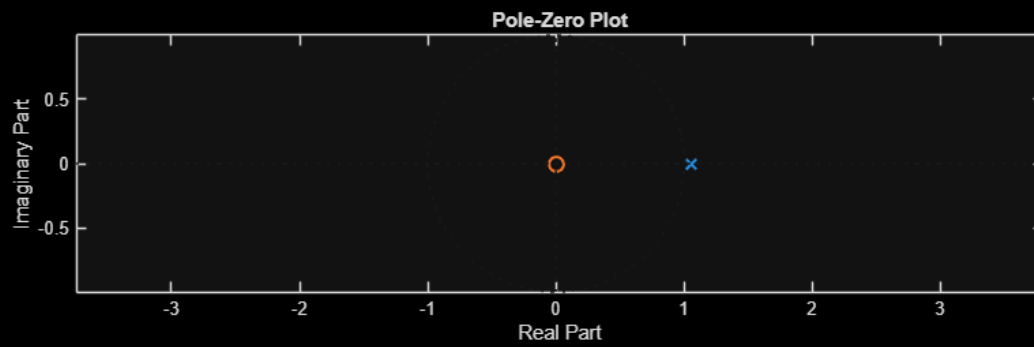
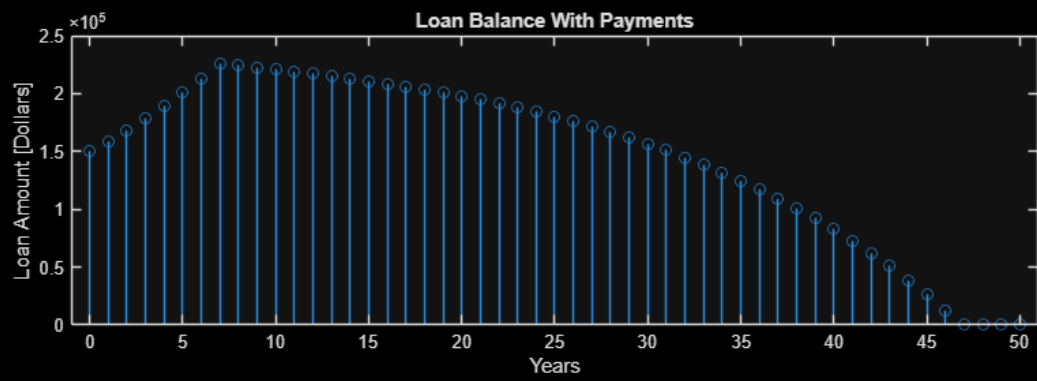
b = zeros(1,Q);
b(1) = 1;
for m = 1:Q-1
    b(m+1) = gamma*(1+alpha)^(m-1);
end

x = zeros(N,1);
x(1) = 150000;

y = filter(b,a,x);

figure
subplot(211)
stem(n,max(y,0))
xlabel('Years')
ylabel('Loan Amount [Dollars]')
title('Loan Balance with Payments Based on Current Loan Value')

subplot(212)
pzplot(b,a)
axis equal
snapnow;
```



3 (g) ANSWER QUESTION

```
% The loan goes to zero.
% After payments begin, the effective recursion is  $y[n] = 0.96 \cdot y[n-1]$ ,
% so the balance decays exponentially.
% The loan is first less than 10% of the original loan (less than $15000)
% at year  $n = 74$ .

% % to verify
% N = 80;
% n = 0:N-1;
%
% alpha = 0.06;
% gamma = 0.1;
% Q = 8;
%
% a = [1 -(1 + alpha - gamma)];
%
% b = zeros(1,Q);
% b(1) = 1;
% for m = 1:Q-1
%     b(m+1) = gamma*(1+alpha)^(m-1);
% end
%
% x = zeros(N,1);
% x(1) = 150000;
%
% y = filter(b,a,x);
%
% figure
% subplot(211)
% stem(n,max(y,0))
% xlabel('Years')
% ylabel('Loan Amount [Dollars]')
% title('Loan Balance with Payments Based on Current Loan Value')
%
% subplot(212)
% pzplot(b,a)
% axis equal
% snapnow;
```

QUESTION 4: LOAN DIFFERENCE EQUATION

4 (a) PLOT OUTPUT AND POLE-ZERO PLOT

4 (b) ANSWER QUESTION

4 (c) PLOT OUTPUT AND POLE-ZERO PLOT

4 (d) ANSWER QUESTION

4 (e) ANSWER QUESTION

4 (f) PLOT OUTPUT AND POLE-ZERO PLOT

4 (g) ANSWER QUESTION

**ALL FUNCTIONS SUPPORTING THIS CODE %
% %**

=====

=====

NOTE: YOU DO NOT NEED TO ADD COMMENTS IN THE CODE BELOW.
WE JUST NEEDED POLE-ZERO PLOTTING CODE AND THUS WROTE IT.

=====

```
function pzplot(b,a)
% PZPLOT(B,A) plots the pole-zero plot for the filter described by
% vectors A and B. The filter is a "Direct Form II Transposed"
% implementation of the standard difference equation:
%
%      a(1)*y(n) = b(1)*x(n) + b(2)*x(n-1) + ... + b(nb+1)*x(n-nb)
%                  - a(2)*y(n-1) - ... - a(na+1)*y(n-na)
%
%
% MODIFY THE POLYNOMIALS TO FIND THE ROOTS
b1 = zeros(max(length(a),length(b)),1); % Need to add zeros to get the
right roots
a1 = zeros(max(length(a),length(b)),1); % Need to add zeros to get the
right roots
b1(1:length(b)) = b;      % New a with all values
a1(1:length(a)) = a;      % New a with all values
% FIND THE ROOTS OF EACH POLYNOMIAL AND PLOT THE LOCATIONS OF THE ROOTS
```

```

h1 = plot(real(roots(a1)), imag(roots(a1)));
hold on;
h2 = plot(real(roots(b1)), imag(roots(b1)));
hold off;

% DRAW THE UNIT CIRCLE
circle(0,0,1)

% MAKE THE POLES AND ZEROS X's AND O's
set(h1, 'LineStyle', 'none', 'Marker', 'x', 'MarkerFaceColor','none',
'linewidth', 1.5, 'markersize', 8);
set(h2, 'LineStyle', 'none', 'Marker', 'o', 'MarkerFaceColor','none',
'linewidth', 1.5, 'markersize', 8);
axis equal;

% DRAW VERTICAL AND HORIZONTAL LINES
xminmax = xlim();
yminmax = ylim();
line([xminmax(1) xminmax(2)], [0 0], 'linestyle', ':', 'linewidth', 0.5,
'color', [1 1 1]*.1)
line([0 0], [yminmax(1) yminmax(2)], 'linestyle', ':', 'linewidth', 0.5,
'color', [1 1 1]*.1)

% ADD LABELS AND TITLE
xlabel('Real Part')
ylabel('Imaginary Part')
title('Pole-Zero Plot')

end

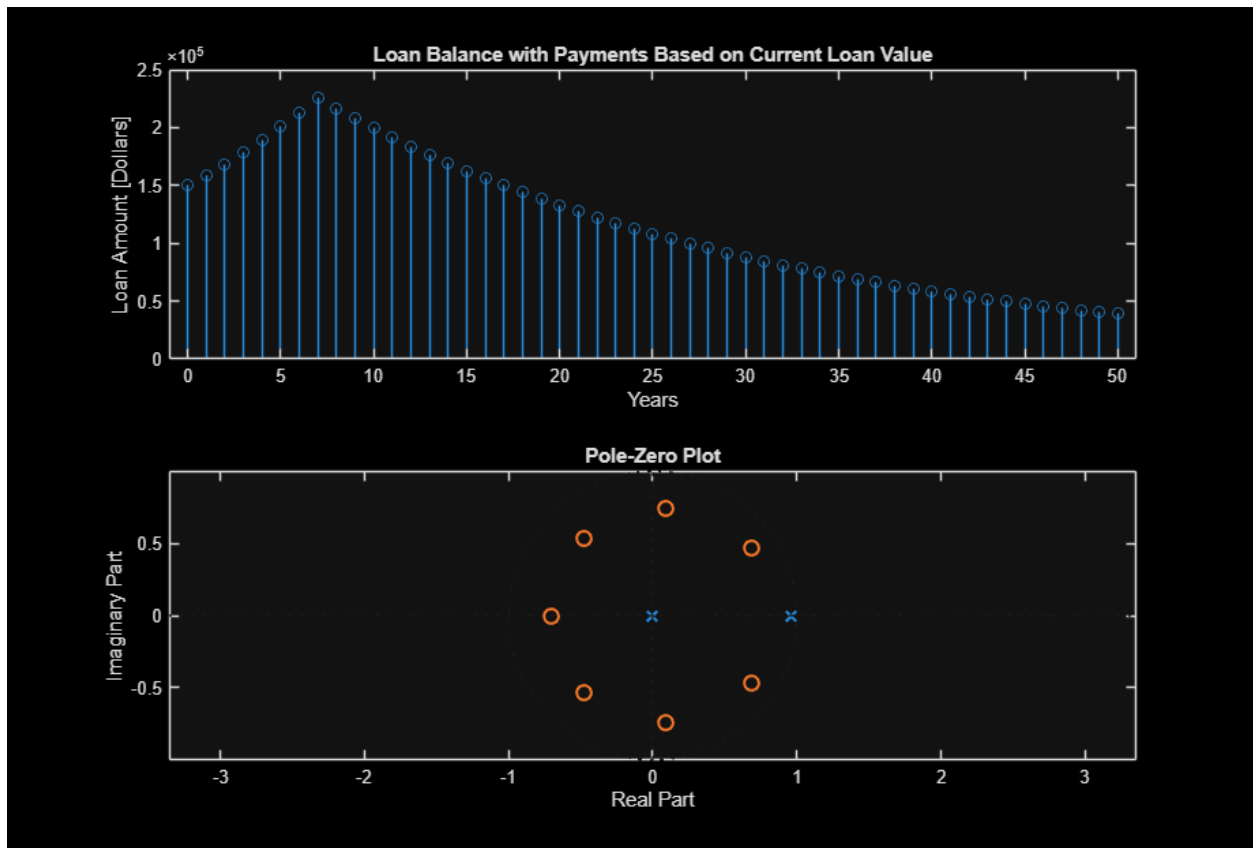
function circle(x,y,r)
% CIRCLE(X,Y,R) draws a circle with horizontal center X, vertical center
% Y, and radius R.
%
% ANGLES TO DRAW
ang=0:0.01:2*pi;

% DEFINE LOCATIONS OF CIRCLE
xp=r*cos(ang);
yp=r*sin(ang);

% PLOT CIRCLE
hold on;
plot(x+xp,y+yp, ':', 'linewidth', 0.5, 'color', [1 1 1]*.1);
hold off;

end

```



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